• Unit 3 • AOS 1 Data Analysis • Lesson 3

Where's the middle? How spread out?

1F Measures of centre and spread: median, quartiles, IQR, mean and standard deviation

CFAT 1.1.1 | Do Now | APK | Learn | Worked examples | You try | Exit tasks

TODAY'S LESSON

Today's lesson

LEARNING INTENTIONS

- [ ] To be able to understand the mean and the median as measures of centre.
- [ ] To be able to understand the range, interquartile range and standard deviation as measures of spread.
- [ ] To be able to know whether to use the median and interquartile range, or the mean and standard deviation, for a particular distribution.
- [ ] To be able to use a CAS calculator to calculate these summary statistics.

LESSON SEQUENCE

1. CFAT 1.1.1 · 20 minutes, 1A–1E
2. Do Now · three current VCAA questions
3. APK · Year 11 skills
4. The median → WE 14, 15, 16
5. Range, quartiles and IQR → WE 17, 18, 19, 20
6. The mean → WE 21
7. Mean or median? Standard deviation · CAS
8. Exit tasks · two VCAA questions
9. Check-in and homework

SUCCESS CRITERIA

By the end of this lesson you'll be able to say...

- ☐ I can find the median value in a data set.
- ☐ I can find the median value from a dot plot.
- ☐ I can find the median value from a stem plot.
- ☐ I can find the range from a stem plot.
- ☐ I can find the IQR from a stem plot when n is even.
- ☐ I can find the IQR from a stem plot when n is odd.
- ☐ I can calculate the mean using the formula.
- ☐ I can calculate the mean and standard deviation using the CAS calculator.

CFAT 1.1.1 · 1A–1E

CFAT 1.1.1: types of data, displays, log scales

HOW IT RUNS

- Individual, in silence
- CAS and your bound reference allowed
- 6 multiple choice + short answer · **13 marks**
- Covers last week: 1A–1E
- Pens down when the timer ends

TIME REMAINING

19:57

Timer running · click on when you're ready to go

Quick write

DO NOW 1 OF 3 · VCAA 2025 NHT · EXAM 1

VCAA: NHT report gives answers only

Review of 1C

The *systolic blood pressure* of 50 people was recorded. The results are summarised in the histogram below.

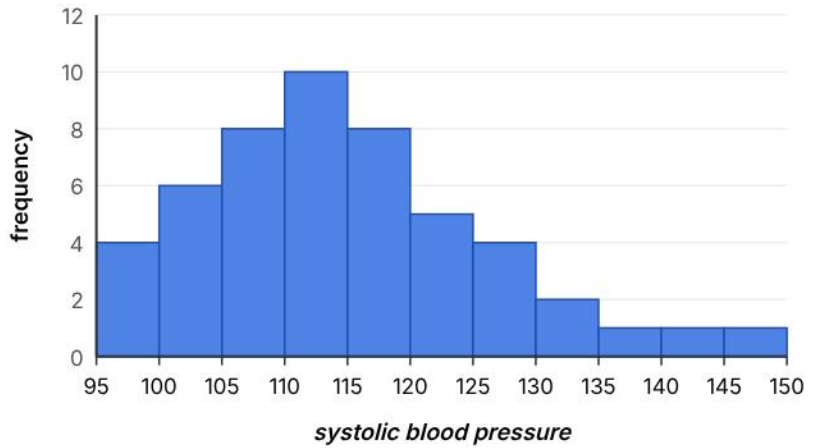
For this group of people, the distribution of their *systolic blood pressure* is best described as

A. symmetric.

B. bell-shaped.

C. **positively skewed.**

D. negatively skewed.



SOLUTION · C

Most values sit at the low end (100–120) and the tail stretches out to the right, towards 150: **positively skewed**.

WHAT'S THE DISTRACTOR?

No % data: the NHT report gives answers only. Watch for D (the tail is on the right, so it is positive).

Multiple choice

DO NOW 2 OF 3 · VCAA 2024 NHT · EXAM 1

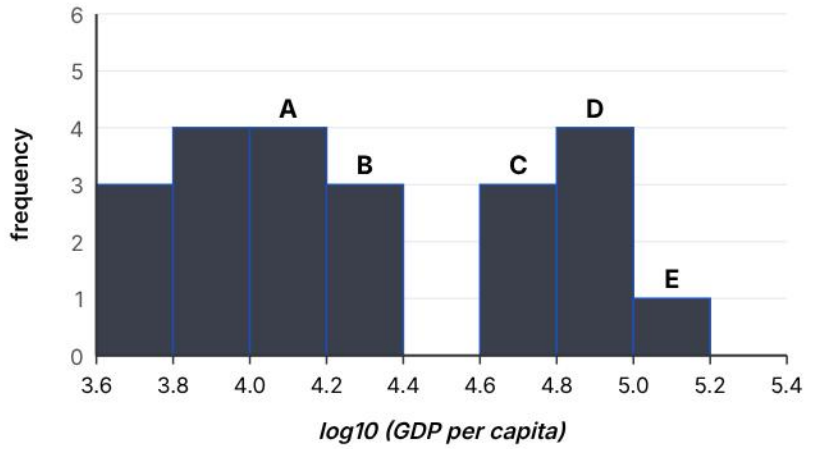
VCAA: NHT report gives answers only

Review of 1E

The histogram below shows the distribution of the *GDP per capita*, in dollars per year, for a sample of 22 countries in 2021 plotted on a logarithmic (base 10) scale.

Australia is one of these 22 countries, and its *GDP per capita* value is $56 281.
Which one of the columns marked A, B, C, D or E would contain Australia's value?

A. A **B.** B **C.** C
D. D **E.** E



SOLUTION · C

$\log_{10}(56\,281) = 4.750\ldots$, which falls in the column from 4.6 to 4.8: column **C**.

WHAT'S THE DISTRACTOR?

No % data: the NHT report gives answers only. Watch for D: 4.75 is close to 4.8, but it is still below it.

Multiple choice

DO NOW 3 OF 3 · VCAA 2024 NHT · EXAM 2

VCAA: NHT report gives answers only

In the year 2021 there were 12 days where a new moon appeared in Sydney Harbour.
For these 12 days the occurrence of the highest high tides (HHT) and lowest low tides (LLT) occurred before, on or after the appearance of the new moon.
These records are displayed in the table below.

Table 3

LLT	on	before	before	after	after	after
	after	after	after	after	after	after
HHT	on	before	before	before	after	after
	after	on	on	after	after	after

Data: http://www.bom.gov.au/australia/tides/

a. Use the data from Table 3 to complete the following frequency table. (2 marks)

Table 4

Occurrence	Frequency	
	LLT	HHT
before	2	3
on	1	3
after	9	6
Total	**12**	**12**

1 **LLT**
Tally both LLT rows: before 2, on 1, after 9.

2 **HHT**
Before 3, on 3, after 6. Each column totals 12.

Quick write

APK · FROM YEAR 11

Back to the party

The ages of a group of people at a party:

19 21 22 56 34 19 27 28 34 17
66 37 20 20 21 45 26 19 23 29

a Determine the mean and median of the ages of the people at the party. Which is the preferred measure of centre?

b Determine the quartiles and hence the IQR of the ages of the people at the party.

c Find the standard deviation of the ages of the people at the party.

age (years) Key: 1 | 7 = 17

1	7 9 9 9
2	0 0 1 1 2 3 6 7 8 9
3	4 4 7
4	5
5	6
6	6

$\bar{x} = \frac{583}{20} = 29.15$; $M = \frac{23+26}{2} = 24.5$. **Median**: skewed with outliers (56, 66).

$Q_1 = \frac{20+20}{2} = 20$, $Q_3 = \frac{34+34}{2} = 34$, IQR = **14**

CAS: $s_x = 13.18\ldots$ → **s ≈ 13.2**

Quick write

LEARN · 1F

Two questions about any distribution

WHERE IS THE CENTRE?

median (M): the middle value

mean ($\bar{x}$): the balance point

HOW SPREAD OUT IS IT?

range: largest − smallest

IQR: spread of the middle 50%

standard deviation (s): spread around the mean

They come in pairs: median with IQR · mean with standard deviation.

Class discussion

LEARN · 1F

The median: the middle value



n = 8 is even: average the 4th and 5th values, M = (5 + 6) ÷ 2 = 5.5

4 values below it, 4 values above it

THE MEDIAN

The median is the middle value in an ordered data set.

For *n* data values the median is located at the $\frac{n+1}{2}$th position. When:

- *n* is odd, the median will be the middle data value
- *n* is even, the median will be the average of the two middle data values.

Class discussion

WORKED EXAMPLE 14 · 1F

▶ Video

Finding the median value in a data set

Order each of the following data sets, locate the median, and then write down its value.

a 2 9 1 8 3 5 3 8 1

b 10 1 3 4 8 6 10 1 2 9

a $n = 9$ (odd): $\frac{9+1}{2} = 5$th value

1 1 2 3 **3** 5 8 8 9

Median = 3

b $n = 10$ (even): average of the 5th and 6th

1 1 2 3 **4** **6** 8 9 10 10

Median $= \frac{4+6}{2} = 5$

1. **Order**
   Write the data set in order.
2. **Locate**
   Odd: the middle value. Even: the two middle values.
3. **Check**
   Equal numbers of values on each side of the median.

Cold calling

YOU TRY · 1F

Finding the median value in a data set

Construct a five-number summary for the following data.

3 5 1 10 8 9 6 3 8 6

In order ($n = 10$):

1 3 3 5 6 6 8 8 9 10

minimum = 1, maximum = 10

median: $\frac{10+1}{2} = 5.5$th position, $\frac{6+6}{2} = 6$

Q_1 = median of the lower half = 3 · Q_3 = median of the upper half = 8

Five-number summary: **1, 3, 6, 8, 10**

Quick write

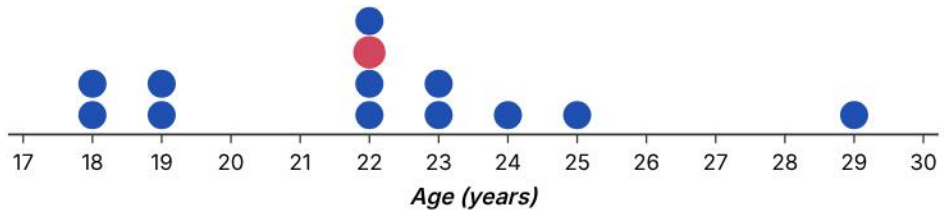
WORKED EXAMPLE 15 · 1F

Video

Finding the median value from a dot plot

The dot plot opposite displays the age distribution (in years) of the 13 members of a local cricket team.

Determine the median age of these cricketers and mark its location on the dot plot.



$n = 13$: the median is the $\frac{13+1}{2} = 7$th value from either end.

Median = **22 years**

1 Count
13 dots, so the 7th value is the middle.

2 Locate
Count in from the left: 18, 18, 19, 19, 22, 22, **22**.

3 Write
Mark it on the plot. Median = 22 years.

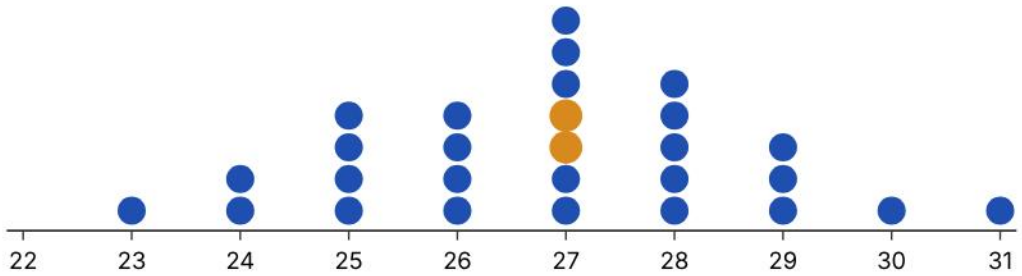
Cold calling

YOU TRY · 1F

Finding the median value from a dot plot

Consider the dot plot.

a. Determine the median.



There are 28 data points: $\frac{28+1}{2} = 14.5$, so average the 14th and 15th data points.

$\frac{27+27}{2} = 27$

Quick write

WORKED EXAMPLE 16 · 1F

▶ Video

Finding the median value from a stem plot

The stem plot opposite displays the maximum temperature (in °C) for 12 days in January.

Determine the median maximum temperature for these 12 days.

Key: 0|8 = 8°C

1	8 9 9
2	0 2 5 7 8 9 9
3	1 3

Count in from both ends: 25 and 27 are the 6th and 7th values.

$M = \frac{25+27}{2} = 26$°C

1 **n is even**
12 values: the median is the average of the two middle values.

2 **Locate**
Count in 6 from each end: 25 and 27.

3 **Average**
$\frac{25+27}{2} = 26$°C

Cold calling

YOU TRY · 1F

Finding the median value from a stem plot

Consider the stem plot.

a. Calculate the median.

> There are 33 data points: $\frac{33+1}{2} = 17$, so the median is the 17th data point.
>
> Median = **485**

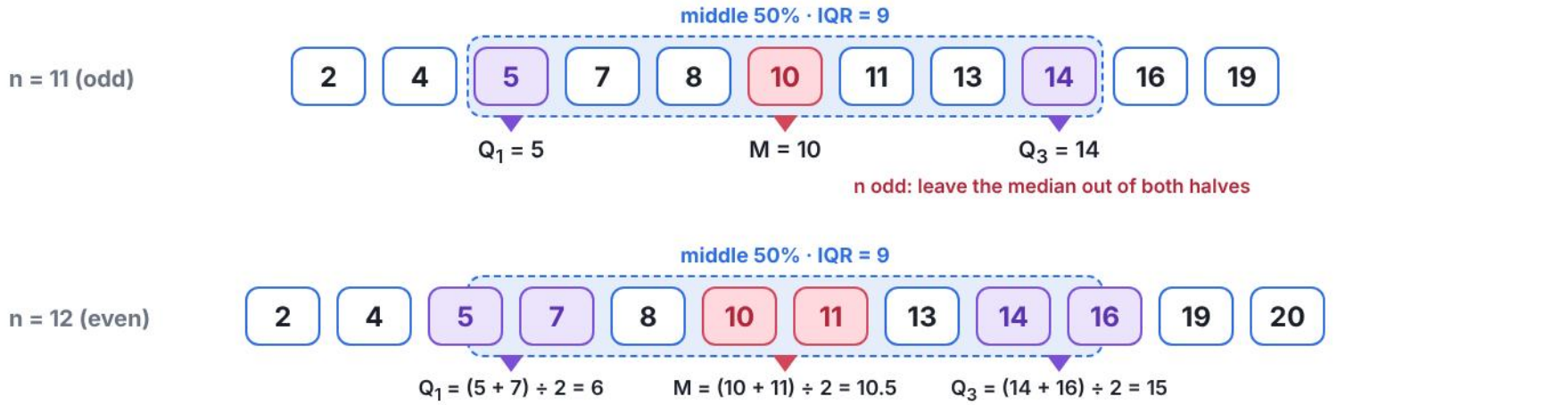
Key: 40 | 7 = 407

40	7
41	
42	
43	
44	4 6
45	1 3 3
46	2 6 7
47	4 7 8 9
48	0 0 4 **5** 6 8
49	1 2 5 5 5 9
50	3 4 4 7 8 8 9 9

Quick write

LEARN · 1F

Quartiles cut the ordered data into quarters



Range R = largest − smallest

Q_1 = median of lower half · Q_3 = median of upper half · IQR = $Q_3 - Q_1$

Class discussion

WORKED EXAMPLE 17 · 1F

▶ Video

Finding the range from a stem plot

The stem plot opposite displays the maximum temperature (in °C) for 12 days in January.

Determine the temperature range over these 12 days.

Key: 0|8 = 8°C

1	8 9 9
2	0 2 5 7 8 9 9
3	1 3

Lowest = 18, highest = 33

range $= 33 - 18 =$ **15°C**

1 Identify

The lowest and highest values in the stem plot.

2 Substitute

$R =$ largest − smallest $= 33 - 18 = 15$°C

Cold calling

YOU TRY · 1F

Finding the range from a stem plot

The following stem plot displays the number of *goals scored* by 16 netball teams over a season.

a. Calculate the range of *goals scored*.

minimum = 22, maximum = 42

range $= 42 - 22 =$ **20 goals**

Key: 2 | 2 = 22 goals

2	2 3 5 7 9
3	0 1 3 5 7 7 8
4	0 1 2 2

Quick write

WORKED EXAMPLE 18 · 1F

▶ Video

Finding the IQR from an ordered stem plot when n is even

Find the interquartile range of the weights of the 18 cats whose weights are displayed in the ordered stem plot below.

Weight (kg) 1 | 2 represents 1.2 kg

1	9
2	1 3 5 **8**
3	0 0 4 9 9
4	0 4 5 **8**
5	0 3
6	3 4

18 values: nine in the lower 'half', nine in the upper 'half'.

Lower half: 1.9 2.1 2.3 2.5 **2.8** 3.0 3.0 3.4 3.9 → Q_1 = **2.8**

Upper half: 3.9 4.0 4.4 4.5 **4.8** 5.0 5.3 6.3 6.4 → Q_3 = **4.8**

$\text{IQR} = Q_3 - Q_1 = 4.8 - 2.8 =$ **2.0**

1 **Split**
18 values: 9 in each half.

2 **Q_1**
5th value from the bottom.

3 **Q_3**
5th value from the top.

4 **IQR**
$Q_3 - Q_1$

Cold calling

YOU TRY · 1F

Finding the IQR from an ordered stem plot when n is even

The following stem plot displays the number of *goals scored* by 16 netball teams over a season.

b. Calculate the IQR of *goals scored*.

Key: 2 | 2 = 22 goals

2	2 3 5 7 9
3	0 1 3 5 7 7 8
4	0 1 2 2

$n = 16$: position of median $= \frac{16+1}{2} = 8.5$, so 8 values in each half.

Lower half: 22 23 25 27 29 30 31 33 → $Q_1 = \frac{27+29}{2} = 28$

Upper half: 35 37 37 38 40 41 42 42 → $Q_3 = \frac{38+40}{2} = 39$

IQR $= 39 - 28 =$ **11 goals**

Quick write

WORKED EXAMPLE 19 · 1F

Video

Finding the IQR from an ordered stem plot when n is odd

The stem plot shows the life expectancy (in years) for 23 countries. Find the IQR for life expectancies.

Stem: 5|2 = 52 years

5	2
5	5 6
6	4
6	6 6 7 9
7	1 2 2 3 3 4 4 4 4
7	5 5 6 6 7 7

23 values: the median is the 12th value from either end, **73**.

n is odd, so the median is left out: 11 values below it and 11 above it.

Q_1 = midpoint of the bottom 11 = **66**

Q_3 = midpoint of the top 11 = **75**

$$\text{IQR} = Q_3 - Q_1 = 75 - 66 = 9$$

Check: 52 55 56 64 66 | **66** | 67 69 71 72 72 | **73** | 73 74 74 74 74 | **75** | 75 76 76 77 77 → four groups of 5 values

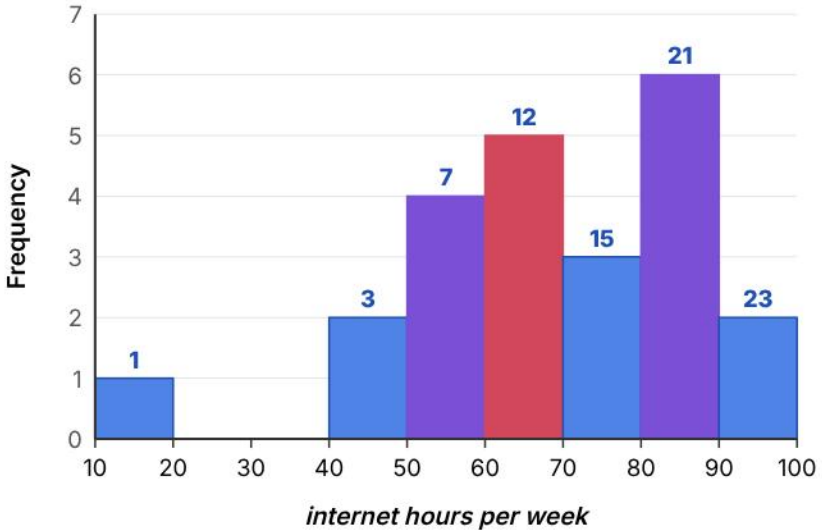
Cold calling

WORKED EXAMPLE 20 · 1F

Video

Finding the median and quartiles from a histogram

The histogram shows the average number of hours per week a group of 23 people spent on the internet. Find possible values for the median and quartiles of this distribution.



Add the frequencies from left to right: 1, 3, 7, 12, ...

M = 12th value from the bottom: median is in the interval **60–70**

Q_1 = 6th value from the bottom: **50–60**

Q_3 = 6th value from the top (2 + 6): **80–90**

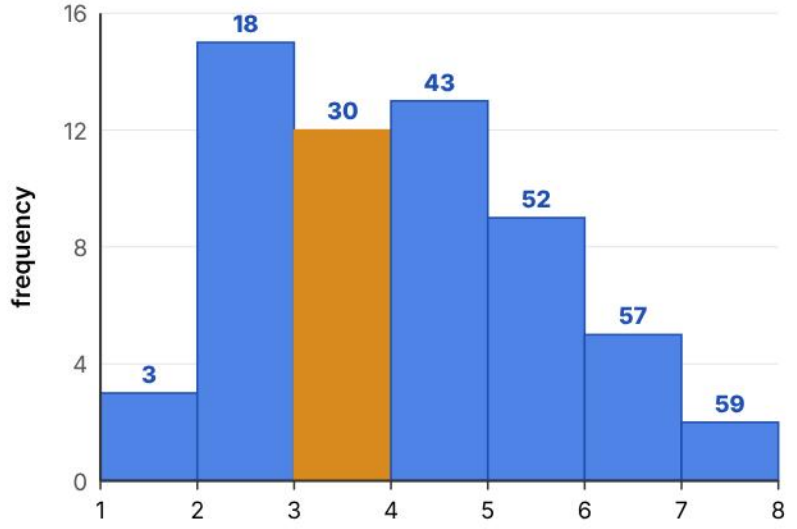
Cold calling

YOU TRY · 1F

Finding the median and quartiles from a histogram

Consider the histogram.

a. Estimate the median.



$3 + 15 + 12 + 13 + 9 + 5 + 2 = 59$ data points; median at $\frac{59+1}{2} = 30$th

Cumulative frequency: 3, 18, 30, ... The 30th data point occurs in the 3–<4 interval.

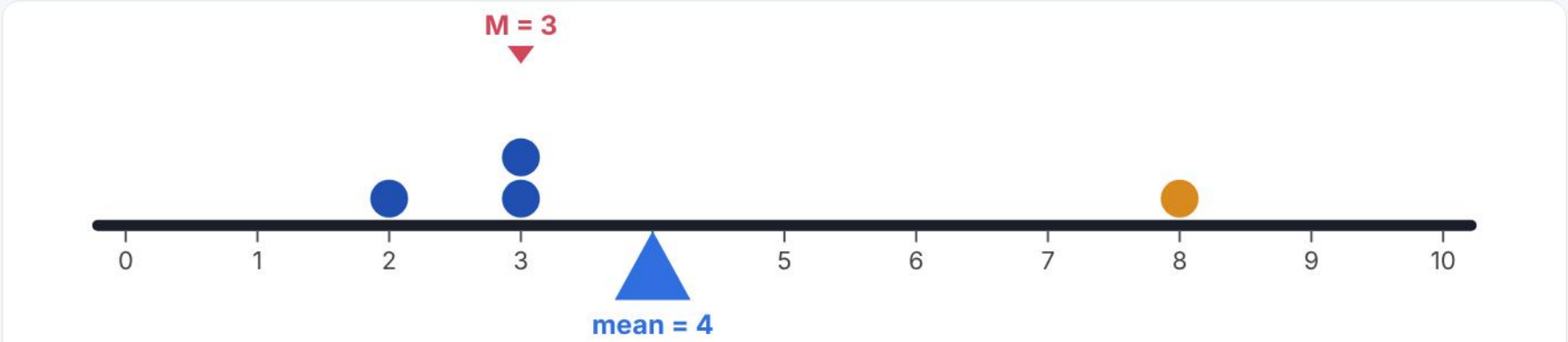
$\frac{3+4}{2}$ =**approximately 3.5**

Quick write

LEARN · 1F

The mean is the balance point

$$\bar{x} = \frac{\sum x}{n}$$ $\sum$ = 'sum of', x = a data value, n = number of values



Data: 2 3 3 8 → mean = 2 + 3 + 3 + 8 ÷ 4 = **4**, median = **3**

Resistant: the mean is affected by changing the largest data value but the median is not. Drag the orange dot to try it.

WORKED EXAMPLE 21 · 1F

▶ Video

Calculating the mean from the formula

The following is a set of reaction times (in milliseconds): 38 36 35 43 46 64 48 25
Write down the values of the following, correct to one decimal place.

a n b $\sum x$ c $\bar{x}$

a n is the number of data values: $n = 8$

b $\sum x = 38 + 36 + 35 + 43 + 46 + 64 + 48 + 25 = 335$

c $\bar{x} = \dfrac{\sum x}{n} = \dfrac{335}{8} = 41.875 \rightarrow \mathbf{41.9}$

1 **Count**
n = how many values.

2 **Add**
$\sum x$ = the sum of the values.

3 **Divide**
$\bar{x} = \sum x \div n$, then round.

Cold calling

YOU TRY · 1F

Calculating the mean from the formula

The amount of money spent, in dollars, of 10 customers at a supermarket is shown. Calculate the mean amount of money spent.

11.60 55.50 7.95 42.15 17.10 2.00 82.55 26.85 5.40 21.90

$$\sum x = 11.60 + 55.50 + 7.95 + \cdots + 21.90 = 273, \quad n = 10$$

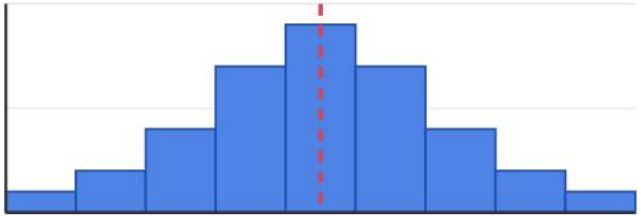
$$\bar{x} = \frac{273}{10} = 27.3 \rightarrow \textbf{\$27.30}$$

Quick write

LEARN · 1F

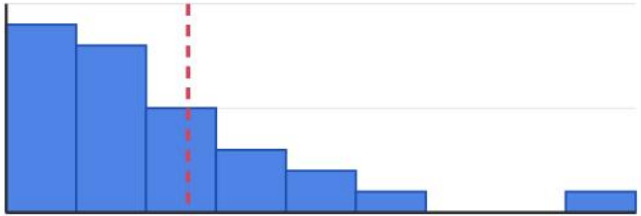
Mean or median?

SYMMETRIC, NO OUTLIERS



mean ≈ median: **either** can be used

CLEARLY SKEWED AND/OR OUTLIERS



the tail drags the mean: use the **median**

CHOOSING BETWEEN THE MEAN AND THE MEDIAN

The mean and the median are both measures of the centre of a distribution. If the distribution is:

- symmetric and there are no outliers, either the mean or the median can be used to indicate the centre of the distribution
- clearly skewed and/or there are outliers, it is more appropriate to use the median to indicate the centre of the distribution.

Key box: Cambridge General Mathematics Units 3&4, 1F p. 52 · Sketches: teacher illustration (dashed line = mean)

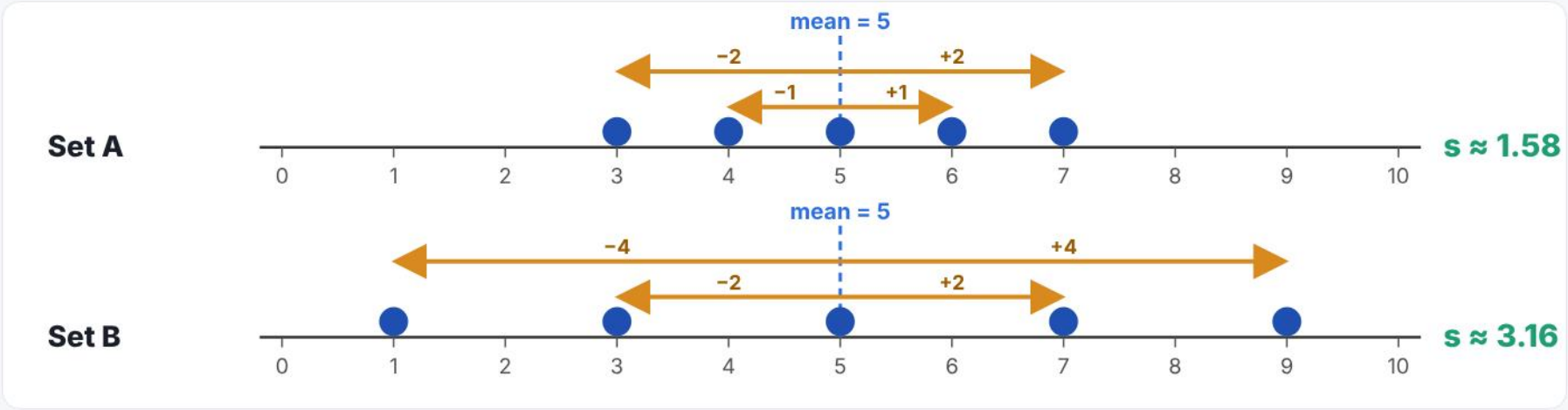
Think, pair, share

LEARN · 1F

Standard deviation: spread around the mean

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Roughly, the typical distance of the data values from the mean. On CAS it is $\mathbf{s_x}$. Same mean, different spread:



Formula: Cambridge General Mathematics Units 3&4, 1F p. 52 · Animation data: teacher example

Class discussion

CAS · 1F

Mean and standard deviation from a list

The following are the heights (in cm) of a group of women. `176 160 163 157 168 172 173 169`

Determine the mean and standard deviation of the women's heights. Give your answers correct to two decimal places.

TI-NSPIRE

1. New document → **Lists & Spreadsheet**. Name a list **height**; enter the data.
2. Add **Calculator**.
3. Menu › Statistics › Stat Calculations › **One-Variable Statistics**. Num of Lists: 1.
4. X1 List: **height**. OK.
5. Read $\bar{x}$ and **sx** (the sample standard deviation).

CLASSPAD

1. **Statistics** app. Name a list **height**; enter the data.
2. Calc › **One-Variable**.
3. XList: main\height · Freq: 1. OK.
4. Read $\bar{x}$ and $\mathbf{s_x}$. Scroll for the median, Q_1, Q_3, min and max.

ANSWER

$$\bar{x} = 167.25 \text{ cm}$$
$$s = 6.67 \text{ cm}$$

Use s_x, not σ_x. General Maths always uses the sample standard deviation.

CAS · 1F

Summary statistics from a frequency table

When values repeat (a dot plot or frequency table), enter each value once and its frequency in a second list. Check with the cricket team from Example 15:

TI-NSPIRE

1. Lists & Spreadsheet: **age** in column A, **freq** in column B.
2. Add Calculator › Menu › Statistics › Stat Calculations › **One-Variable Statistics**.
3. X1 List: **age** · Frequency List: **freq**. OK.
4. Read $\bar{x}$, **sx**, **n**, MedianX.

CLASSPAD

1. Statistics app: **age** in list1, **freq** in list2.
2. Calc › **One-Variable**.
3. XList: main\age · Freq: main\freq. OK.
4. Read $\bar{x}$, $\mathbf{s_x}$, **n**, Med.

age	18	19	22	23	24	25	29
freq	2	2	4	2	1	1	1

CHECK

$n = 13 \cdot \bar{x} = 22 \cdot s_x = 3.08 \cdot$ median $= 22$ ✓

Check n. If n is not the total frequency, the frequency list was not selected.

Reference: ClassPad Help 402 (statistical calcs from a frequency table) · Data: Cambridge Example 15

Class discussion

YOU TRY · CAS

Calculating the mean and standard deviation using the CAS calculator

The *weight*, in kilograms, of a sample of 10 rugby players are shown. Determine the mean and standard deviation of the rugby players' *weight*, rounded to two decimal places.

80 95 85 91 102 93 87 78 84 90

One-Variable Statistics on the list **weight**.

mean $\bar{x} =$ **88.50 kg**

standard deviation $s_x = 7.230\ldots \rightarrow$ **7.23 kg**

Quick write

EXIT TASK 1 OF 2 · EDROLO 1G Q11 (ADAPTED FROM VCAA 2020 EXAM 2 Q4AII)

VCAA: **77%** scored 1/1

Show what you know

The *body density*, in kilograms per litre, and *weight*, in kilograms, of a sample of 12 orangutans are shown in the table.

body density **(kg/litre)**	1.07	1.07	1.08	1.08	1.03	1.05	1.07	1.06	1.07	1.09	1.02	1.09
weight **(kg)**	70.1	90.4	73.2	85.0	84.3	95.6	71.7	95.0	80.2	87.4	94.9	65.3

For these 12 orangutans, determine the mean of their *body density*, in kilograms per litre. (1 mark)

SOLUTION

$$\bar{x} = \frac{12.78}{12} = \mathbf{1.065}\ \text{kg/litre}$$

Examiner: the required answer was 1.065; rounding did not apply, so 1.07 was not accepted.

EXIT TASK 2 OF 2 · VCAA 2026 NHT EXAM 2 Q1B, D

VCAA: NHT report gives answers only

The table below displays the typical body weight, in kilograms, for a sample of 19 different mammals.

b. Complete the five-number summary in the table below for the variable *body weight*, in kilograms, by writing in the values for the minimum and the median. (2 marks)

Minimum	Q_1	Median	Q_3	Maximum
2.5	4.3	35	100	250

d. The mean body weight is 67 kg, rounded to the nearest whole number.
Explain why the median is a better measure of the centre of the body weight data than the mean. (1 mark)

> b. Minimum = 2.5 (rabbit). n = 19, so the median is the $\frac{19+1}{2} = 10$th value: **35** (kangaroo).
>
> d. The data is (positively) skewed and/or has an outlier, so the mean (67) is dragged above most values; the median is not affected.

Mammal	Body weight (kg)
okapi	250
gorilla	210
pig	190
Brazilian tapir	160
jaguar	100
grey seal	85
giant armadillo	60
sheep	56
chimpanzee	52
kangaroo	35
goat	28
roedeer	15
baboon	11
rhesus monkey	6.8
raccoon	4.3
red fox	4.2
cat	3.3
echidna	3.0
rabbit	2.5

Data: Adapted from T Allison and DV Cicchetti, 'Sleep in mammals: ecological and constitutional correlates', Science, vol. 194, no. 4266, 12 November 1976, pp. 732–734 <lib.stat.cmu.edu/datasets/sleep>

Quick write

CHECK-IN

How did you go?

- ☐ I can find the median value in a data set.
- ☐ I can find the median value from a dot plot.
- ☐ I can find the median value from a stem plot.
- ☐ I can find the range from a stem plot.
- ☐ I can find the IQR from a stem plot when n is even.
- ☐ I can find the IQR from a stem plot when n is odd.
- ☐ I can calculate the mean using the formula.
- ☐ I can calculate the mean and standard deviation using the CAS calculator.

Thumbs up, sideways or down for each one, then rate it on your worksheet: ✓ got it · ~ nearly · × not yet



True or false

HOMEWORK

Homework

CAMBRIDGE · EXERCISE 1F (PART 1)

Median from data	**Q1–2**
Median from a dot plot	**Q3**
Median and range from a stem plot	**Q4**
Median and quartiles from a dot plot	**Q5**
Median and quartiles from a stem plot	**Q6–9**
Median and quartiles from a histogram	**Q10–11**

CAMBRIDGE · EXERCISE 1F (PART 2)

Mean, median and mode from data	**Q12–17**
The concept of standard deviation	**Q18–20**
Mean and standard deviation on CAS	**Q21–22**
Exam 1 style questions	**Q23–26**

REVISION BOOK

The **1F** section